

Representation Theory

Labix

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Abstract

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1 Basic Definitions of Representations

1.1 Representation of an Algebra

Definition 1.1.1 (Representation of an Algebra)

Let R be a commutative ring. Let A be an R -algebra. A representation of A is a left R -module M and a homomorphism of R -algebras

$$\rho : A \rightarrow \text{End}_R(M)$$

Let A, M be sets. Recall that there is a natural bijection

$$\text{Hom}_{\text{Sets}}(A \times M, M) \cong \text{Hom}_{\text{Sets}}(A, \text{Hom}_{\text{Sets}}(M, M))$$

given by curling. A representation of an R -algebra A is an element in

$$\text{Hom}_{\text{Alg}_R}(A, \text{End}_R(M)) \subseteq \text{Hom}_{\text{Sets}}(A, \text{Hom}_{\text{Sets}}(M, M))$$

Therefore we may ask what kinds of maps of the form $A \times M \rightarrow M$ give a representation of A .

Proposition 1.1.2

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $f : A \times M \rightarrow M$ be a function. Then the map $a \mapsto f(a, -)$ defines a representation of A if and only if the following are true.

- f is an R -module homomorphism.
- f defines an A -module structure on M .

The A -module structure can also be seen in the representation $\rho : A \rightarrow \text{End}_R(M)$. It is given by $a \cdot m = \rho(a)(m)$. In particular, not every A -module M gives a representation; the action of A on M must be bilinear in the scalars R .

Definition 1.1.3 (Subrepresentation)

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $\rho : A \rightarrow \text{End}_R(M)$ be a representation of A . A subrepresentation of ρ is an R -submodule $N \leq M$ such that the map $\rho|_N : A \rightarrow \text{End}_R(N)$ defined by $\rho|_N = \rho(a)|_N$ is a representation of A .

Definition 1.1.4 (Quotient Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $\rho : A \rightarrow \text{End}_R(M)$ be a representation of A . Let $N \leq M$ be a subrepresentation of M . Define the quotient representation of M by N to be the representation $A \rightarrow \text{End}_R(M/N)$ given by

$$a \mapsto (m + N \mapsto \rho(a)(m) + N)$$

We need to check that the quotient is indeed a representation. Firstly, the map is well defined since N is a subrepresentation, so that for $m \in N$ we know that $\rho(a)(m) \in N$.

Definition 1.1.5 (Sum of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Define the sum of ρ and τ to be the representation $\rho \oplus \tau : A \rightarrow \text{End}_R(M \oplus N)$ given by

$$(\rho \oplus \tau)(a)(m, n) = (\rho(a)(m), \tau(a)(n))$$

Since representations of an algebra A are A -modules, it follows that the following notions in Rings and Modules apply to representations.

- A representation is simple (irreducible) if it is a simple A -module (equivalently, it has no non-trivial subrepresentations).
- A representation is semisimple if it is a direct sum of simple representations.

Definition 1.1.6 (Indecomposable Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $\rho : A \rightarrow \text{End}_R(M)$ be a representation of A . We say that M is indecomposable if M is not the direct sum of representations.

Lemma 1.1.7

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $\rho : A \rightarrow \text{End}_R(M)$ be a representation of A . If ρ is irreducible, then ρ is indecomposable.

The converse need not be true in general.

Example 1.1.8

Let k be a field. Define a representation $\rho : k[x] \rightarrow \text{End}_k(k^2) \cong M_2(k)$ by $x \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ then extended it R -linearly and respecting products of x . Then ρ is indecomposable but reducible.

Proof

Suppose for a contradiction that ρ is decomposable. Then there exists 1-dimensional linear subspaces L_1, L_2 of k^2 such that ρ_{L_1} and ρ_{L_2} are representations. Suppose that $L_1 = k\langle v_1 \rangle$ and $L_2 = k\langle v_2 \rangle$. Then $x \cdot v_1 \in L_1$ and $x \cdot v_2 \in L_2$ implies that v_1 and v_2 are linearly independent eigenvectors of $\rho(x)$. But by an easy computation, we see that the only eigenvalue of $\rho(x)$ is 1 and its eigenspace has dimension 1. Hence this is a contradiction.

From the above, we see that the eigenspace of $\rho(x)$ is a subrepresentation of ρ . Hence ρ is reducible. ■

1.2 Maps between Representations

Definition 1.2.1 (Homomorphisms of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . A homomorphism from ρ to τ is a function $\phi : M \rightarrow N$ such that the following are true.

- ϕ is an R -module homomorphism.
- The following diagram commutes:

$$\begin{array}{ccc} A \times M & \xrightarrow{\rho} & M \\ \text{id}_A \times \phi \downarrow & & \downarrow \phi \\ A \times N & \xrightarrow{\tau} & N \end{array}$$

Since M and N are A -modules, the definition of homomorphism implies that such a morphism must be an A -module homomorphism.

Definition 1.2.2 (Isomorphisms of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. We say that ϕ is an isomorphism if there exists a homomorphism of representations $\psi : N \rightarrow M$ such that

$$\psi \circ \phi = \text{id}_M \quad \text{and} \quad \phi \circ \psi = \text{id}_N$$

Lemma 1.2.3

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. Then ϕ is an isomorphism if and only if ϕ is an isomorphism of R -modules.

Theorem 1.2.4 (First Isomorphism Theorem for Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. Then the following are true.

- $\ker(\phi)$ is a subrepresentation of M .
- $\text{im}(\phi)$ is a subrepresentation of N .
- There is an isomorphism of representations

$$M / \ker(\phi) \cong \text{im}(\phi)$$

given by ϕ .

Proposition 1.2.5

Let R be a commutative ring. Let A be a finite R -algebra. Let $\rho : A \rightarrow \text{End}_R(M)$ be an irreducible representation of A . Then M is finitely generated.

Proof

Let $0 \neq m \in M$. Then $\rho(A)(m)$ is a submodule of M . Since M is irreducible, we must have $\rho(A)(m) = M$. Since A is finite over R , we must have $\rho(A)(m)$ is finitely generated over R . Hence M is finitely generated over R . ■

1.3 Schur's Lemma

Recall from Rings and Modules that we have Schur's lemma: A homomorphism $M \rightarrow N$ of simple R -modules is either the 0 map or an isomorphism.

Lemma 1.3.1 (Schur's Lemma III)

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a finite dimensional k -vector space. Let $\rho : A \rightarrow \text{End}_k(V)$ be an irreducible representations of A . Let $\phi : V \rightarrow V$ be a homomorphism of representations. Then $\phi = \lambda \text{id}_V$ for some $\lambda \in k$.

Proof

Since ϕ is a homomorphism of simple A -modules, by Schur's lemma we know that $\phi = 0$ or ϕ is an isomorphism. If $\phi = 0$ then we are done. Otherwise, since k is algebraically closed, the endomorphism ϕ admits an eigenvalue $\lambda \in k$ and eigenvector $v \in V$. Consider the homomorphism of representations $\phi - \lambda \text{id}_V$. By Schur's lemma, this map is either 0 or an isomorphism. But it is not an isomorphism since $\phi(v) - \lambda v = 0$. Hence $\phi - \lambda \text{id}_V = 0$. So we are done. ■

Corollary 1.3.2

Let k be an algebraically closed field. Let A be a commutative k -algebra. Let V be a finite dimensional k -vector space. Let $\rho : A \rightarrow \text{End}_k(V)$ be an irreducible representations of A . Then $\dim(V) = 1$.

Proof

By Schur's lemma III, we know that $\rho(a) : V \rightarrow V$ is a scalar operator for all $a \in A$. This means that $\rho(a)(v) = \lambda_a v$ for some $\lambda_a \in k$ and all $v \in V$. For any fixed $v_0 \in V$, this implies that $k\langle v_0 \rangle$ is a subrepresentation of ρ . If $\dim(V) \geq 2$, then $k\langle v_0 \rangle$ is a non-trivial subrepresentation, contradicting the irreducible assumption. Hence $\dim(V) = 1$. ■

In other words, every finite dimensional representation of a commutative algebra over an algebraically closed field must be of dimension 1. And these are easy to classify.

1.4 Representations of Finite Algebras over an Algebraically Closed Field

Proposition 1.4.1

Let k be an algebraically closed field. Let A be a finite k -algebra. Then there are only finitely many irreducible representations of A up to isomorphism.

Proposition 1.4.2

Let k be an algebraically closed field. Let A be a finite k -algebra. Then the following are equivalent.

- A is semisimple.
- Every representation of A is semisimple.
- $\dim(A) = \sum_{i=1}^n \dim(V_i)^2$ where $V_1, \dots, V_n$ is the complete list of non-isomorphic irreducible representations of A .

1.5 Character Theory

Let k be an algebraically closed field. Recall that the trace of a k -linear map between finite dimensional vector spaces have a basis independent description.

Definition 1.5.1 (Characters of a Representation)

Let k be an algebraically closed field. Let A be a k -algebra. Let $\rho : A \rightarrow \text{End}_k(V)$ be a finite dimensional representation of A . Define the character of ρ to be function $\chi_V : A \rightarrow k$ given by

$$\chi_V(a) = \text{tr}(\rho(a))$$

Definition 1.5.2 (Characters of an Algebra)

Let k be an algebraically closed field. Let A be a k -algebra. Let $\phi : A \rightarrow k$ be a function. We say that ϕ is a character of A if there exists a finite dimensional representation V of A such that $\phi = \chi_V$.

Proposition 1.5.3

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be isomorphic finite dimensional representations of A . Then $\chi_V = \chi_W$.

Lemma 1.5.4

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a finite dimensional representation of A . Let $W \leq V$ be a subrepresentation. Then we have

$$\chi_V = \chi_W + \chi_{V/W}$$

Proposition 1.5.5

Let k be an algebraically closed field. Let A be a k -algebra. Let $\rho : A \rightarrow \text{End}_k(V)$ be a finite dimensional representation of A . Then ρ descends to a well defined function $\rho : \frac{A}{[A, A]} \rightarrow k$.

Proposition 1.5.6

Let k be an algebraically closed field. Let A be a k -algebra. If A is semisimple, then the set

$$\{\chi_{[V]} \mid V \text{ is a finite dimensional representation of } A\}$$

is a basis for $\text{Hom}_k\left(\frac{A}{[A, A]}, k\right)$.

2 Representation Theory of Groups

2.1 Representations of a Group

Definition 2.1.1 (Representations of a Group)

Let G be a group. Let k be a field. A representation of G is a k -vector space V together with a group homomorphism $\rho : G \rightarrow \text{End}_k(V)$.

By the natural isomorphism

$$\text{Hom}_{\text{Alg}_k}(k[G], \text{End}_k(V)) \cong \text{Hom}_{\text{Grp}}(G, \text{End}_k(V))$$

we see that representations of a group is the same as representation of the group ring.

Example 2.1.2

The following are representations of $D_8 = \langle r, s \mid r^4, s^2, rsrs \rangle$.

- $\rho : D_8 \rightarrow GL_2(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- $\rho : D_8 \rightarrow GL_n(\mathbb{C})$ defined by $\rho(r) = \rho(s) = I_n$.
- $\rho : D_8 \rightarrow \mathbb{C}^\times$ defined by $\rho(r) = \pm 1$ and $\rho(s) = \pm 1$.
- $\rho : D_8 \rightarrow GL_4(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

Definition 2.1.3 (The Regular Representation of a Group)

Let G be a group. Let k be a field. Define the regular representation $\text{reg} : G \rightarrow \text{End}_k(k[G])$ of G on k by the formula

$$\text{reg}(g) \left(\sum_{h \in G} \lambda_h h \right) = \sum_{h \in G} \lambda_h gh$$

Definition 2.1.4 (Permutation Representations)

Let X be a set. Let G be a group acting on X . Let k be a field. Define the permutation representation $\rho : G \rightarrow GL(k\langle X \rangle)$ of G on X over k by the formula

$$\rho(g)(x) = g \cdot x$$

and then extending it k -linearly.

Every group G has an action on itself by left multiplication. In this case the permutation representation is just the regular representation.

2.2 The Space of Homomorphisms between Representations

Proposition 2.2.1

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V, W be irreducible $k[G]$ -modules. Then we have

$$\dim_k(\text{Hom}_{k[G]}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Proof

If V is not isomorphic to W , then by Schur's lemma we must have $\text{Hom}_{k[G]}(V, W) = \{0\}$. Suppose that $V \cong W$. Let $\varphi : V \rightarrow W$ be a $k[G]$ -module isomorphism. Let $\psi \in \text{Hom}_{k[G]}(V, W)$. Then we have $\varphi^{-1} \circ \psi : V \rightarrow V$ is a $k[G]$ -module homomorphism. Then $\varphi^{-1} \circ \psi = \lambda \text{id}_V$ by a corollary of Schur's lemma. Hence $\psi = \lambda \varphi$ so that $k\langle \varphi \rangle = \text{Hom}_{k[G]}(V, W)$. ■

Proposition 2.2.2

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V, W be $k[G]$ -modules. If $\text{Hom}_{k[G]}(V, W) \neq \{0\}$, then there exists $k[G]$ -submodules $V' \leq V$ and $W' \leq W$ such that $V' \cong W'$.

Proposition 2.2.3

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V_i, W_j be irreducible $k[G]$ -modules for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then we have

$$\dim_k \left(\text{Hom}_{k[G]} \left(\bigoplus_{i=1}^n V_i, \bigoplus_{j=1}^m W_j \right) \right) = \sum_{i=1}^n \sum_{j=1}^m \dim_k (\text{Hom}_{k[G]}(V_i, W_j))$$

Proposition 2.2.4

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V be a $k[G]$ -module. Then we have

$$\dim_k (\text{Hom}_{k[G]}(k[G], V)) = \dim(V)$$

Proof

Let $v_1, \dots, v_{\dim(V)} \in V$ be a basis for V . Define $\varphi_i : k[G] \rightarrow V$ by $\varphi_i(g) = g \cdot v_i$ for $g \in G$ and extend it k -linearly. It is clearly a $k[G]$ -module homomorphism. Hence $\varphi_i \in \text{Hom}_{k[G]}(k[G], V)$. I claim that $\varphi_1, \dots, \varphi_{\dim(V)}$ is a basis for $\text{Hom}_{k[G]}(k[G], V)$. Let $\psi \in \text{Hom}_{k[G]}(k[G], V)$. Then

$$\psi(1) = \sum_{i=1}^{\dim(V)} \lambda_i v_i = \sum_{i=1}^{\dim(V)} \lambda_i \varphi_i(1)$$

for some $\lambda_i \in k$. Now since ψ is a $k[G]$ -module homomorphism, we have

$$\psi(a) = a\psi(1) = a \sum_{i=1}^{\dim(V)} \lambda_i \varphi_i(1) = \sum_{i=1}^{\dim(V)} \lambda_i \varphi_i(a)$$

for any $a \in k[G]$. Hence $\varphi_1, \dots, \varphi_{\dim(V)}$ spans $\text{Hom}_{k[G]}(k[G], V)$. Now suppose that $\sum_{i=1}^{\dim(V)} \lambda_i \varphi_i = 0$ for some $\lambda_i \in k$. Evaluating both sides at $1 \in k[G]$ gives $\sum_{i=1}^{\dim(V)} \lambda_i v_i = 0$. But since $v_1, \dots, v_{\dim(V)}$ is a basis for V , we conclude that $\lambda_1 = \dots = \lambda_{\dim(V)} = 0$. Hence $v_1, \dots, v_{\dim(V)}$ are linearly independent. ■

2.3 Structural Results of Representations of Finite Groups

From section 1 and from Rings and Modules, we know a few facts about representations of a finite group G over k .

- Every representation of G is finite dimensional over k .
- Morphisms between irreducible representations of G are scalar operators.
- If G is abelian, then every irreducible representation of G has dimension 1.
- Maschke's theorem says that if $\text{char}(k)$ does not divide $|G|$, then every representation of G is a direct sum of irreducible representations of G .

Definition 2.3.1 (Composition Factors)

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V be $k[G]$ -a module. Define the composition factors of V to be irreducible $k[G]$ -modules $V_1, \dots, V_n$ such that

$$V \cong \bigoplus_{i=1}^n V_i$$

Denote the set of isomorphism classes of composition factors of V to be

$$\text{Comp}(V) = \{[V_1], \dots, [V_n]\}$$

Definition 2.3.2 (Multiplicity of an Irreducible Submodule)

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V, W be $k[G]$ -modules such that W is irreducible. Define the multiplicity of W in V to be

$$\text{mult}_V(W) = \{[V_i] \in \text{Comp}(V) \mid W \in [V_i]\}$$

Example 2.3.3

The representation of $C_3 = \langle x \mid x^3 \rangle$ on $\mathbb{F}_3^2$ given by

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not semisimple.

Corollary 2.3.4

Let G be a finite group. Let k be a field such that $\text{char}(k)$ does not divide $|G|$. Let V be an irreducible $k[G]$ -module. Then V is a composition factor of $k[G]$.

Proof

Let $0 \neq v \in V$. Define a function $T : k[G] \rightarrow V$ by

$$T\left(\sum_{g \in G} \lambda_g g\right) = \sum_{g \in G} \lambda_g g v$$

where $k[G]$ is the regular representation. Clearly it is a homomorphism of representations. Since $v \neq 0$ and V is irreducible, we must have $\text{im}(T) = V$. By Maschke's theorem, we have $k[G] = \ker(T) \oplus W$ for some subrepresentation W of $k[G]$. By the first isomorphism theorem, we have $W \cong \text{im}(T) = V$ so that V is a subrepresentation of $k[G]$. ■

Corollary 2.3.5

Let G be a finite group. Let k be a field such that $\text{char}(k)$ does not divide $|G|$. Then there are only finitely many non-isomorphic irreducible $k[G]$ -modules.

Corollary 2.3.6

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let V, W be $k[G]$ -modules such that W is irreducible. Then we have

$$\dim_k(\text{Hom}_{k[G]}(V, W)) = \text{mult}_V(W)$$

Let V be an irreducible $k[G]$ -module. Combining the above proposition and corollary, we deduce that

$$\text{mult}_{k[G]}(V) = \dim(V)$$

Theorem 2.3.7

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $k[G]$ -modules. Then we have

$$|G| = \sum_{i=1}^n \dim(V_i)^2$$

Example 2.3.8

Let G be a group of order 8. Let k be an algebraically closed field such that $\text{char}(k) \neq 2$. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $k[G]$ -modules. Then exactly one of the following is true (up to reordering of the modules).

- $n = 8$ and $\dim(V_i) = 1$.
- $n = 5$ and $\dim(V_1) = \dots = \dim(V_4) = 1$ and $\dim(V_5) = 2$.

2.4 Characters of Finite Groups

Similar to the case of representations, the character of a representation of a finite group $\chi : k[G] \rightarrow k$ is the same information as a function $\chi : G \rightarrow k$ simply by restriction and extension via scalars.

Proposition 2.4.1

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let $\rho : G \rightarrow \text{End}_k(V)$ be a representation of G . Then the following are true.

- $\chi_V(1_G) = \dim(V)$.
- $\chi_V(hgh^{-1}) = \chi(g)$ for all $g, h \in G$.

The second item implies that χ descends to a function on conjugation classes $\chi : \text{Cl}(G) \rightarrow k$.

Proposition 2.4.2

Let G be a finite group. Let $\rho : G \rightarrow \text{End}_{\mathbb{C}}(V)$ be a complex representation of G . Then the following are true.

- Let $g \in G$ be an element of order n . Then we have

$$\chi_V(g) = \sum_{i=0}^{n-1} e^{2\pi i/n}$$

- Let $g \in G$. Then we have $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Proof

Let $H = \langle g \rangle$. Then V is also a $\mathbb{C}[H]$ -module. Let $V = \bigoplus_{i=1}^m V_i$ be the decomposition of V into irreducible $\mathbb{C}[H]$ -submodules. Since H is cyclic, we must have $\dim(V_i) = 1$. ■

Let G be a finite group. Since $\mathbb{C}[G]$ is a semisimple $\mathbb{C}$ -algebra, we know that the $\mathbb{C}$ -vector space

$$\text{Hom}_{\mathbb{C}}\left(\frac{G}{[G, G]}, \mathbb{C}\right) = \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$$

has a basis given by the character of G . We now define an inner product of the $\mathbb{C}$ -space and show that the characters form an orthogonal basis.

Definition 2.4.3 (The Inner Product Space of Class Functions)

Let G be a finite group. Define the inner product space of class functions of G to be the $\mathbb{C}$ -vector space

$$\text{Hom}_k\left(\frac{G}{[G, G]}, k\right) = \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$$

together with the inner product given by

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

Proposition 2.4.4

Let G be a finite group. Let ϕ, ψ be characters of G . Then we have $\langle \phi, \psi \rangle \in \mathbb{R}$.

Proof

Let ϕ, ψ be characters. Then we have

$$\begin{aligned}
 \overline{\phi, \psi} &= \frac{1}{|G|} \sum_{g \in G} \phi(g) \overline{\psi(g)} \\
 &= \frac{1}{|G|} \sum_{g \in G} \overline{\phi(g)} \psi(g) \\
 &= \frac{1}{|G|} \sum_{g \in G} \phi(g^{-1}) \psi(g) \\
 &= \frac{1}{|G|} \sum_{h \in G} \phi(h) \psi(h^{-1}) && (\text{Let } h = g^{-1}) \\
 &= \langle \phi, \psi \rangle
 \end{aligned}$$

so we are done. ■

Proposition 2.4.5

Let G be a finite group. Let V, W be irreducible $\mathbb{C}[G]$ -modules. Then we have

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Theorem 2.4.6**2.5 Orthogonality Relations of Characters**

Definition 2.5.1 (Set of Functions from Group to $\mathbb{C}$) Let G be a finite group. Denote

$$\mathbb{C}[G] = \{ \phi : G \rightarrow \mathbb{C} \mid \phi \text{ is a map of sets} \}$$

the set of all functions from G to $\mathbb{C}$.

Lemma 2.5.2 Let V be a finite dimensional irreducible $\mathbb{C}G$ -module. Let $f : V \rightarrow V$ be a G -linear map. Define

$$\tilde{f}(v) = \frac{1}{|G|} \sum_{g \in G} g \cdot (f(g^{-1} \cdot v))$$

Then we must have that

$$\tilde{f} = \frac{\text{tr}(f)}{\dim(V)} I_V$$

Proof It is clear by the averaging trick that $\tilde{f}$ is a G -linear map. By Schur's lemma III, we

conclude that $\tilde{f} = \lambda I_V$ for some λ . Now we have that

$$\begin{aligned} \operatorname{tr} \left(\frac{1}{|G|} \sum_{g \in G} g(f(g^{-1})) \right) &= \operatorname{tr}(\lambda I_V) \\ \frac{1}{|G|} \sum_{g \in G} \operatorname{tr}(g(f(g^{-1}))) &= \operatorname{tr}(\lambda I_V) \\ \frac{1}{|G|} \sum_{g \in G} \operatorname{tr}(f) &= \lambda \cdot \dim(V) \\ \operatorname{tr}(f) &= \lambda \cdot \dim(V) \end{aligned}$$

We conclude that $\tilde{f} = \lambda I_V = \frac{\operatorname{tr}(f)}{\dim(V)} I_V$. ■

Theorem 2.5.3 Let G be a finite group. Then $\mathbb{C}[G]$ is an inner product space over $\mathbb{C}$ where the Hermitian product $\langle \cdot, \cdot \rangle : \mathbb{C}[G] \times \mathbb{C}[G] \rightarrow \mathbb{C}$ is defined by

$$\langle \phi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \phi(g) \overline{\psi(g)}$$

Moreover, $\dim_{\mathbb{C}}(\mathbb{C}[G]) = |G|$.

Proof It is clear that $\langle -, - \rangle$ is linear in the first variable and anti-linear in the second variable. We also have that

$$\overline{\langle \phi, \psi \rangle} = \overline{\frac{1}{|G|} \sum_{g \in G} \phi(g) \overline{\psi(g)}} = \frac{1}{|G|} \sum_{g \in G} \psi(g) \overline{\phi(g)} = \langle \psi, \phi \rangle$$

We also have that

$$\langle \phi, \phi \rangle = \frac{1}{|G|} \sum_{g \in G} |\phi(g)|^2 > 0$$

Thus $\mathbb{C}[G]$ is an inner product space.

I claim that a basis of $\mathbb{C}[G]$ is given by $\delta_g : G \rightarrow \mathbb{C}$ which is the Kronecker-delta function. It is clear that they are linearly independent for varying g because if $\sum_{g \in G} \lambda_g \delta_g = 0$, then applying g to both sides gives $\lambda_g = 0$. Moreover, any $\psi : G \rightarrow \mathbb{C}$ assigns each $g \in G$ to a complex number $\psi(g)$. Then

$$\psi = \sum_{g \in G} \psi(g) \delta_g$$

shows that ψ can be decomposed into linear combinations of the Kronecker-delta functions. ■

In particular, since $\dim_{\mathbb{C}}(\mathbb{C}[G])$ is $|G|$ and $\mathbb{C}G$ is also a vector space of dimension $|G|$, this means that

$$\mathbb{C}[G] \cong \mathbb{C}G$$

Theorem 2.5.4 Let U, V be finite dimensional $\mathbb{C}G$ -modules. Then

$$\langle \chi_U, \chi_V \rangle = \begin{cases} 1 & \text{if } U \sim V \\ 0 & \text{otherwise} \end{cases}$$

Moreover, $U \sim V$ if and only if $\chi_U = \chi_V$.

Proof We have that

$$\begin{aligned}
 \langle \chi_U, \chi_V \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_U(g) \chi_V(g^{-1}) \\
 &= \frac{1}{|G|} \sum_{g \in G} \left(\sum_i (\rho_U(g))_{ii} \sum_j (\rho_V(g^{-1}))_{jj} \right) \\
 &= \sum_{i,j} \left(\frac{1}{|G|} \sum_{g \in G} \rho_U(g)_{ii} \rho_V(g^{-1})_{jj} \right) \\
 &= \sum_{i,j} \left(\frac{1}{|G|} \sum_{g \in G} e_i^T \rho_U(g)_{ii} e_i e_j^T \rho_V(g^{-1})_{jj} e_j \right) \\
 &= \sum_{i,j} e_i^T \left(\frac{1}{|G|} \sum_{g \in G} \rho_U(g)_{ii} E_{i,j} \rho_V(g^{-1})_{jj} \right) e_j
 \end{aligned}$$

Let $\widetilde{E}_{i,j} = \frac{1}{|G|} \sum_{g \in G} \rho_U(g)_{ii} E_{i,j} \rho_V(g^{-1})_{jj}$. Now $\widetilde{E}_{i,j}$ is a linear transformation $V \rightarrow U$ and hence lies in $\text{Hom}(V, U)$. Moreover, $g \cdot \widetilde{E}_{i,j} = \widetilde{E}_{i,j}$ hence $\widetilde{E}_{i,j} \in \text{Hom}_G(V, U)$.

If U is not similar to V , then by Schur's lemma the G -morphism $\widetilde{E}_{i,j} : V \rightarrow U$ is 0. If $U \sim V$, then $\chi_U = \chi_V$. Then it suffices to prove the case for when $U = V$ by lemma 1.2.6. Then we have that $\widetilde{E}_{i,j}$ is an isomorphism by Schur's lemma and hence can only be a diagonal matrix since it is a G -morphism. Then we have that

$$\begin{aligned}
 \sum_{i,j} e_i^T \widetilde{E}_{i,j} e_j &= \sum_i e_i^T \widetilde{E}_{i,i} e_i \\
 &= \text{tr}(\widetilde{E}_{i,j}) \\
 &= \dim(V) \frac{1}{\dim(V)} \text{tr}(E_{i,i}) \\
 &= 1
 \end{aligned} \tag{2.3.2}$$

It remains to show that $U \sim V$ if and only if $\chi_U = \chi_V$. If $U \sim V$, then 2.2.2 implies that U and V have the same characteristic. Now suppose that U and V have the same characteristic. Let $W_1, \dots, W_k$ be a complete list of pairwise non-isomorphic irreducible $\mathbb{C}G$ -modules. By Maschke's theorem, we have that $U \sim \bigoplus_{i=1}^k (W_i)^{\oplus n_i}$ and $V \sim \bigoplus_{i=1}^k (W_i)^{\oplus m_i}$. By lemma 2.2.6, we have that $\chi_U = \sum_{i=1}^k n_i \chi_{W_i}$ and $\chi_V = \sum_{i=1}^k m_i \chi_{W_i}$. By the above theorem, the χ_{W_i} are linearly independent. By assumption, $\chi_U = \chi_V$ implies that $n_i = m_i$. Hence we conclude that

$$U \sim \bigoplus_{i=1}^k (W_i)^{\oplus n_i} = \bigoplus_{i=1}^k (W_i)^{\oplus m_i} \sim V$$

and so we are done. ■

This shows that χ is a complete invariant for $\mathbb{C}G$ -modules.

Lemma 2.5.5 Let U be a finite dimensional $\mathbb{C}G$ -module. Then U is irreducible if and only if $\langle \chi_U, \chi_U \rangle = 1$.

Proof Suppose that U is irreducible. Then trivially $U \sim U$ hence by 2.3.4 we conclude that $\langle \chi_U, \chi_U \rangle = 1$. Now suppose that $\langle \chi_U, \chi_U \rangle = 1$ and let $U \sim \bigoplus_{i=1}^k (W_i)^{\oplus n_i}$ where $W_1, \dots, W_k$ is a complete list of pairwise non-isomorphic irreducibles. Then $\chi_U = \sum_{i=1}^k n_i \chi_{W_i}$. We have that

$$\begin{aligned} \langle \chi_U, \chi_U \rangle &= \left\langle \sum_{i=1}^k n_i \chi_{W_i}, \sum_{i=1}^k n_i \chi_{W_i} \right\rangle \\ &= \sum_{i,j=1}^k n_i n_j \langle \chi_{W_i}, \chi_{W_j} \rangle \\ &= \sum_{i=1}^k (n_i)^2 \end{aligned}$$

By assumption, this is equal to 1. But the sum is strictly positive and hence there exists j such that $n_j = 1$ and $n_i = 0$ for all $i \neq j$. Hence $U \sim W_j$ and so U is irreducible. $\blacksquare$

2.6 Multiplicity and the Isotypic Decomposition

Definition 2.6.1 (Multiplicity) Let G be a finite group. Let V be a non-trivial finite dimensional $\mathbb{C}G$ -module. Suppose that V decomposes into $V = \bigoplus_{i=1}^r U_i$ where each U_i is irreducible. Let U be an irreducible $\mathbb{C}G$ -module. Define the multiplicity of U in V as

$$\text{mult}_U(V) = |\{U_i \mid U \cong U_i\}|$$

Intuitively multiplicity means that the number of isomorphic copies of U lying inside V , for each irreducible U .

Lemma 2.6.2 Let G be a finite group. Let V be a non-trivial finite dimensional $\mathbb{C}G$ -module. Let U be an irreducible $\mathbb{C}G$ -module. Then we have that

$$\text{mult}_U(V) = \langle \chi_U, \chi_V \rangle$$

Proof Let $V = \bigoplus_{i=1}^r U_i$ where each U_i is irreducible. Let $W_1, \dots, W_k$ be a complete list of pairwise non-isomorphic irreducibles. Then each U_i is isomorphic to W_j for some $1 \leq j \leq k$. We have that

$$\chi_V = \sum_{i=1}^r \chi_{U_i} = \sum_{j=1}^k n_j \chi_{W_j}$$

where $n_j = \text{mult}_{W_j}(V)$. We have that

$$\begin{aligned} \langle \chi_U, \chi_V \rangle &= \left\langle \chi_U, \sum_{j=1}^k n_j \chi_{W_j} \right\rangle \\ &= \sum_{j=1}^k n_j \langle \chi_U, \chi_{W_j} \rangle \end{aligned}$$

Using 2.3.4, the terms in the sum is non-zero if and only if U is isomorphic to W_j . Each $W_1, \dots, W_k$ are distinct and hence U is only isomorphic to exactly one of W_j . In this case the sum becomes $n_j = \text{mult}_{W_j}(V) = \text{mult}_U(V)$. $\blacksquare$

Thus given a decomposition

$$U \sim \bigoplus_{i=1}^k (W_i)^{\oplus n_i}$$

of a $\mathbb{C}G$ -module into a complete list of irreducible $\mathbb{C}G$ -modules, $\text{mult}_{W_i}(U)$ is simply the exponent n_i in the decomposition. In particular, the above lemma says that

$$n_i = \langle \chi_U, \chi_{W_i} \rangle$$

Lemma 2.6.3 Let V be a finite dimensional $\mathbb{C}G$ -module. Let $W_1, \dots, W_k$ be the complete list of pairwise non-isomorphic irreducible $\mathbb{C}G$ -submodules of V . Then

$$\sum_{i=1}^k (\dim(W_i))^2 = |G|$$

Recall the Artin-Wedderburn theorem from Rings and Modules, which states that if R a ring considered as a left R -module that is semisimple, then one can decompose R into the direct product of matrix rings over division rings. The following theorem applies the result to $\mathbb{C}G$ considered as a $\mathbb{C}G$ -module over itself. We will give an independent proof of the result.

Theorem 2.6.4 (Result of Artin-Wedderburn Theorem) Let G be a finite group. Let V be a finite dimensional $\mathbb{C}G$ -module. Let $W_1, \dots, W_k$ be the complete list of pairwise non-isomorphic irreducible $\mathbb{C}G$ -submodules of V . Let

$$f : \mathbb{C}G \rightarrow \text{End}(W_1) \times \dots \times \text{End}(W_k)$$

be defined by $f(g) = (\rho_{W_1}(g), \dots, \rho_{W_k}(g))$ and extended linearly. Then f is a $\mathbb{C}$ -algebra isomorphism.

Proof It is clear that f is a G -homomorphism. Moreover, f sends identity to identity. Write $W = \text{End}(W_1) \times \dots \times \text{End}(W_k)$. Now I claim that f is a ring homomorphism. This is because

$$\begin{aligned} f \left(\left(\sum_{g \in G} \alpha_g g \right) \left(\sum_{h \in G} \beta_h h \right) \right) &= f \left(\sum_{g, h \in G} \alpha_g \beta_h gh \right) \\ &= \sum_{g, h \in G} \alpha_g \beta_h f(gh) \\ &= \sum_{g, h \in G} \alpha_g \beta_h (\rho_W(gh) f(1)) \\ &= \sum_{g, h \in G} \alpha_g \beta_h (\rho_W(gh) 1_W) \\ &= \sum_{g, h \in G} \alpha_g \beta_h (\rho_W(g) 1_W) \cdot_W (\rho_W(h) 1_W) \\ &= \left(\sum_{g \in G} \alpha_g \rho_W(g) 1_W \right) \cdot_W \left(\sum_{h \in G} \beta_h \rho_W(h) 1_W \right) \\ &\quad (W \text{ is a } \mathbb{C}G\text{-module}) \\ &= \left(\sum_{g \in G} \alpha_g \rho_W(g) f(1) \right) \cdot_W \left(\sum_{h \in G} \beta_h \rho_W(h) f(1) \right) \\ &= \left(\sum_{g \in G} \alpha_g f(g \cdot 1) \right) \cdot_W \left(\sum_{h \in G} \beta_h f(h \cdot 1) \right) \\ &= f \left(\sum_{g \in G} \alpha_g g \right) \cdot_W f \left(\sum_{h \in G} \beta_h h \right) \end{aligned}$$

Thus f is a $\mathbb{C}$ -algebra homomorphism. By the above lemma, we have that

$$\begin{aligned}\dim(\mathbb{C}G) &= |G| \\ &= \sum_{i=1}^k (\dim(W_i))^2 \\ &= \dim(W)\end{aligned}$$

Hence we just have to show that f is injective. Suppose that $a = \sum_{g \in G} \alpha_g g \in \ker(f)$. Then $\rho_{W_i}(a) = 0$ for all i . Therefore $\rho_{W_i}(a)(w) = 0$ for all $w \in W$. Since $W_1, \dots, W_k$ is a complete list of pairwise non-isomorphic irreducibles, by Maschke's theorem, for every V a $\mathbb{C}G$ -module, we must have $\rho_V(a) = 0$. In particular, for $V = \mathbb{C}G$ we have that $a \cdot_{\mathbb{C}G} b = 0$ for all $b \in \mathbb{C}G$. By choosing $b = 1$, we conclude that $a = 0$. Thus $\ker(f) = 0$ and so we conclude. $\blacksquare$

Proposition 2.6.5 Let G be a group. Denote Cl_G the set of conjugacy classes in G . Then

$$\dim(Z(\mathbb{C}G)) = |\text{Cl}_G|$$

Theorem 2.6.6 Let G be a finite group. Then

$$|\hat{G}| = |\text{Cl}_G|$$

Moreover, the characters of the irreducible representations form an orthonormal basis of the vector space $\mathbb{C}[\text{Cl}_G]$.

Proof By 2.4.4, there is an isomorphism $\mathbb{C}G \cong \text{End}_{W_1} \times \dots \times \text{End}_{W_k}$. Now since $Z(\text{End}_{W_i}) = \mathbb{C}I_{W_i}$, we have that

$$Z(\text{End}_{W_1}) \times \dots \times Z(\text{End}_{W_k}) = Z(\mathbb{C}G)$$

Thus $Z(\mathbb{C}G) = k$. By 2.4.5, we conclude that $k = |\text{Cl}_G|$. $\blacksquare$

Definition 2.6.7 (Isotypic Components) Let V be a finite dimensional $\mathbb{C}G$ -module. Let W_l be an irreducible representation of G . We call the spaces

$$V_l = \bigoplus_{j=1}^{\text{mult}_{W_l}(V)} U_{l,j}$$

given above where $U_{l,j} \cong W_l$ the isotypic components of V .

A representation is said to be isotypic if it contains only one non-zero isotypic component.

While the decomposition of V into irreducible subrepresentations is not unique, the isotypic decomposition is unique up to reordering the summands.

Theorem 2.6.8 Let $W_1, \dots, W_k$ be a complete list of pairwise nonisomorphic irreducible representations of G . For $1 \leq i \leq k$, let

$$a_i = \frac{\dim(W)}{|G|} \sum_{g \in G} \overline{\chi_{W_i}(g)} g \in \mathbb{C}G$$

Let V be a finite dimensional $\mathbb{C}G$ -module. Consider the decomposition into irreducibles:

$$V = \bigoplus_{l=1}^k \bigoplus_{j=1}^{\text{mult}_{W_l}(V)} U_{l,j}$$

with each $U_{l,j} \cong W_l$. Then $\rho_V(a_i) \in \text{End}(V)$ is the projection onto the i th isotypic component of V . In particular, the space V_i is independent of the finer decomposition of V into the direct sum of the $U_{l,j}$.

2.7 Character Tables

Definition 2.7.1 (Character Tables) Let G be a finite group. The character table of G is a table

G	$\text{Cl}_G(g_1)$	$\text{Cl}_G(g_2)$	$\cdots$
χ_1			
χ_2			
$\vdots$			

where the rows are the irreducible characters and the columns are the conjugacy classes of G .

Recall that the set of irreducible characters of a finite group G forms the basis for the vector space $\mathbb{C}[\text{Cl}_G]$ of all class functions. Moreover, the size of such a basis is equal to $|\text{Cl}_G|$. This means that the character table is a square. Moreover, using the orthonormal property of the basis of irreducible characters, we can deduce a number of consequences which can be easily visualized in the character table.

Conversely, we can also use these orthonormal properties to deduce the character table of a group. For instance, a matrix has orthonormal columns if and only if it has orthonormal rows. This is very powerful because the characters is a complete invariant for representations.

Definition 2.7.2 (Augmented Character Tables) Let G be a finite group. The augmented character table of G is a character table of G with each column $\text{Cl}_G(g)$ multiplied by $\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}}$.

Proposition 2.7.3 Let G be a finite group. Write the elements of the augmented character table of G into a square matrix A . Then A is orthonormal.

Proof If χ_1 and χ_2 are two irreducible characters, the inner product of the two rows of A is

$$\begin{aligned} \sum_{\text{Cl}_G(g) \in \text{Cl}_G} \left(\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}} \chi_1(g) \right) \left(\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}} \overline{\chi_2(g)} \right) &= \frac{1}{|G|} \sum_{\text{Cl}_G(g) \in \text{Cl}_G} (|\text{Cl}_G(g)| \chi_1(g) \overline{\chi_2(g)}) \\ &= \frac{1}{|G|} \sum_{g \in G} (\chi_1(g) \overline{\chi_2(g)}) \\ &= \langle \chi_1, \chi_2 \rangle \quad (\chi \text{ is a class function}) \end{aligned}$$

If $\chi_1 \neq \chi_2$, then by 2.3.4 we conclude that the sum is 0 and hence the rows of A are orthogonal. If $\chi_1 = \chi_2$ then by 2.3.4 we conclude that the sum is 1 and hence the rows of A are orthonormal. Thus we are done. ■

Corollary 2.7.4 Let G be a finite group. Then the following are true regarding the character table of G .

- The rows of the character table are orthogonal. This means that

$$\sum_{\text{Cl}_G(g) \in \text{Cl}_G} \chi_1(g) \chi_2(g) = 0$$

for any irreducible characters χ_1 and χ_2 .

- The columns of the character table are orthogonal. This means that if g_1 and g_2 are not in the same conjugacy classes then

$$\sum_{\chi \text{ is irr.}} \chi(g_1) \overline{\chi(g_2)} = 0$$

where the sum is over all irreducible characters.

Proof Write the character table of G into a square matrix A and write the augmented character table of G into a square matrix B . Then B is obtained from A by scaling the columns of A . But B is orthonormal and scaling the rows does not change orthogonality hence A is still an orthogonal matrix (but no longer orthonormal). ■

Corollary 2.7.5 Let G be a finite group and $g \in G$. Then we have

$$\sum_{\chi \text{ is irr.}} \chi(g) \overline{\chi(g)} = \frac{|G|}{|\text{Cl}_G(g)|}$$

where the sum is over all irreducible characters.

Proof Write the augmented character table of G into a square matrix A . Then A is orthonormal. This means that

$$\sum_{\chi \text{ is irr.}} \left(\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}} \chi(g) \right) \left(\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}} \overline{\chi(g)} \right) = 1$$

Thus we have that

$$\begin{aligned} \sum_{\chi \text{ is irr.}} \left(\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}} \chi(g) \right) \left(\frac{\sqrt{|\text{Cl}_G(g)|}}{\sqrt{|G|}} \overline{\chi(g)} \right) &= 1 \\ \frac{|\text{Cl}_G(g)|}{|G|} \sum_{\chi \text{ is irr.}} \chi(g) \overline{\chi(g)} &= 1 \\ \sum_{\chi \text{ is irr.}} \chi(g) \overline{\chi(g)} &= \frac{|G|}{|\text{Cl}_G(g)|} \end{aligned}$$

and so we conclude. ■

3 Induced Representations

3.1 Restricted and Induced Representations

Definition 3.1.1 (Restriction of Representations) Let $H \leq G$ be a subgroup. Let V be a finite dimensional $\mathbb{C}G$ -module. Then H acts on V and we denote the corresponding $\mathbb{C}H$ -module by

$$\text{Res}_H^G V : H \rightarrow GL(V)$$

We write the restriction of the characters as

$$\text{Res}_H^G \chi_V = \chi_{\text{Res}_H^G V} : H \rightarrow GL(V)$$

Note that if V is an irreducible $\mathbb{C}G$ -module, $\text{Res}_H^G V$ may not be irreducible.

Definition 3.1.2 (Induced Representation) Let H be a subgroup of G of index l . Let $g_1H, \dots, g_kH$ represent all the distinct cosets of H in G . Let $\rho : H \rightarrow GL(n, \mathbb{C})$ be a representation. Define the induced representation

$$\text{Ind}_H^G \rho : G \rightarrow GL\left(\bigoplus_{i=1}^k g_i V\right)$$

where each $g_i V$ is isomorphic to V via the map $g_i v \mapsto v$, by the following formula. For $g \in G$, the map $\rho(g) : \bigoplus_{i=1}^k g_i V \rightarrow \bigoplus_{i=1}^k g_i V$ is given by the formula

$$\left(\sum_{i=1}^k g_i v_i\right) \mapsto \left(\sum_{i=1}^k (gg_i) v_i\right)$$

We need to show that the matrix on the right is invertible for all g in order for $\text{Ind}_H^G \rho$ to formally be called as a representation. This is the content of the following theorem.

Proposition 3.1.3 Let G, H be groups. Let V be a $\mathbb{C}G$ -module and let W be a $\mathbb{C}H$ -module. Then there is an isomorphism

$$\text{Hom}_{\mathbb{C}H}(W, \text{Res}_H^G V) = \text{Hom}_{\mathbb{C}G}(\text{Ind}_H^G W, V)$$

Explicitly, this means that any $\mathbb{C}H$ -module homomorphism $W \rightarrow V$ extends uniquely to a $\mathbb{C}G$ -homomorphism $\text{Ind}_H^G W \rightarrow V$.

Theorem 3.1.4 (Frobenius Reciprocity) Let G be a finite group and let H be a subgroup of G . Let χ be a character of G and let ρ be a character of H . Then

$$\langle \text{Ind}_H^G \psi, \chi \rangle_G = \langle \psi, \text{Res}_H^G \chi \rangle_H$$

Proposition 3.1.5 Let G be a finite group and H a subgroup of G of index n . Let $1 : H \rightarrow GL(1, \mathbb{C})$ be the trivial 1-dimensional representation. Then the induced representation $\text{Ind}_H^G 1 : G \rightarrow \text{End}(\mathbb{C}^n)$ is equivalent to the representation $\rho : G \rightarrow GL(\mathbb{C}\mathcal{H})$.

Proof We want to show that $\text{Ind}_H^G 1(g) = \rho(g)$ for any $g \in G$. Since $\rho(g)$ acts by permutation on the basis, the matrix $\rho(g)$ only consists of 0 and 1. Similarly, $\text{Ind}_H^G 1(g)$ by definition has entries

either 0 or 1. Now we have that

$$\begin{aligned}\mathrm{Ind}_H^G 1(g)_{i,j} = 1 &\iff t_i^{-1}gt_j \in H \\ &\iff gt_jH = t_iH \\ &\iff \rho(g)_{i,j} = 1\end{aligned}$$

Thus we conclude. ■

Definition 3.1.6 (The Coset Module) Let G be a finite group and let H be a subgroup of G . Write $\mathcal{H}$ the set of cosets of H . Define the coset module to be the permutation representation $\rho : G \rightarrow \mathrm{GL}(\mathbb{C}\mathcal{H})$ where $\rho(g) : \mathbb{C}\mathcal{H} \rightarrow \mathbb{C}\mathcal{H}$ is defined on the basis by sending t_iH to gt_iH .

Notice that since G acts on G/H , then similar to the case of regular representations, $\mathbb{C}\mathcal{H}$ becomes a $\mathbb{C}G$ -module and hence a representation.

Theorem 3.1.7 Suppose that $\mathcal{H} = \{t_1H, \dots, t_lH\}$ and $\mathcal{H}' = \{s_1H, \dots, s_lH\}$ are two representations of the set of cosets of H in G . Then the two representations constructed from $\mathcal{H}$ and $\mathcal{H}'$ are isomorphic.

Lemma 3.1.8 Let G be a finite group and let H be a subgroup of G . Let $\rho : H \rightarrow \mathrm{GL}(V)$ be a representation with character χ . Then for all $g \in G$, we have that

$$\mathrm{Ind}_H^G \chi(g) = \frac{1}{|H|} \sum_{x \in G} \chi_V(x^{-1}gx)$$

where $\chi(g) = 0$ if $g \notin H$.

3.2 Decomposition of Regular Representations

4 Computations of Representations

4.1 Representations of Abelian Groups

Let $\rho : G \rightarrow \text{GL}(V)$ be a representation. Then for an arbitrary choice of $g \in G$, $\rho(g) : V \rightarrow V$ is necessarily a linear map but may not be a G -linear map. Indeed, $\rho(g) : V \rightarrow V$ is a G -linear map if and only if for all $h \in G$, we have

$$\begin{aligned}\rho(g)(h \cdot v) &= h \cdot \rho(g)(v) \\ \rho(g)(\rho(h)(v)) &= \rho(h)(\rho(g)(v)) \\ \rho(gh)(v) &= \rho(hg)(v)\end{aligned}$$

for all $v \in V$. Thus $g \in Z(G)$. When G is abelian, we can show that if V is irreducible then V is one dimensional over $\mathbb{C}$.

Theorem 4.1.1 Let G be an abelian. If V is an irreducible representation of G over $\mathbb{C}$, then $\dim(V) = 1$.

Proof Assume that V is irreducible. Then $\rho(g) : V \rightarrow V$ is a G -linear map. Indeed we have that

$$\begin{aligned}\rho(g)(h \cdot v) &= \rho(g)(\rho(h)(v)) \\ &= \rho(gh)(v) \\ &= \rho(hg)(v) \\ &= h \cdot (\rho(g)(v))\end{aligned}$$

Thus $\rho(g) : V \rightarrow V$ is a G -linear map. by Schur's lemma III, we conclude that $\rho(g) = \lambda v$ for some eigenvector λ of $\rho(g)$. This means that every subspace of V is G -invariant. Since V is irreducible, $\dim(V) = 1$. ■

4.2 Representations of the Cyclic Group

Since C_n is an abelian group, the irreducible representations of C_n are all 1-dimensional. Let us compute their character tables.

Theorem 4.2.1 Denote $C_n = \langle x \rangle$ the cyclic group. The set of all complex irreducible representations (up to equivalence) of C_n are precisely

$$\{\phi_k : C_n \rightarrow \mathbb{C}^* \mid k = 0, \dots, n-1\}$$

defined as follows. For each $k \in \{0, \dots, n-1\}$, the multiplication map $\phi_k(x^m) : \mathbb{C} \rightarrow \mathbb{C}$ is defined by

$$\phi_k(x^m)(z) = e^{\frac{2\pi i k m}{n}} z$$

Proof Since C_n is abelian, we only need to find the 1-dimensional complex representations. Suppose that $\phi : C_n \rightarrow \mathbb{C}^*$ is a complex representation. Now $\phi(x)$ is an invertible complex number. But

$$1 = \phi(1) = \phi(x^n) = \phi(x)^n$$

implies that $\phi(x)$ must be an n th root of unity. There are in fact n of them. Thus we have obtained n representations $\phi_k : C_n \rightarrow \mathbb{C}^*$ defined by

$$\phi_k(x)(z) = e^{\frac{2\pi i k}{n}} z$$

They are in fact distinct from each other. If ϕ_k is equivalent to ϕ_j , then there exists $x \in \text{GL}(1, \mathbb{C}) = \mathbb{C}^*$ such that

$$x^{-1} e^{\frac{2\pi i k m}{n}} x = e^{\frac{2\pi i j m}{n}}$$

This implies that $e^{\frac{2\pi i k m}{n}} = e^{\frac{2\pi i j m}{n}}$ and hence $k \equiv j \pmod{n}$. Thus we conclude. ■

Theorem 4.2.2 The character table of C_n is given as follows.

G	1	x	$\dots$	x^u	$\dots$	x^{n-1}
Trivial	1	1	$\dots$	1	$\dots$	1
χ_{ϕ_1}	1	$e^{\frac{2\pi i}{n}}$	$\dots$	$e^{\frac{2\pi i u}{n}}$	$\dots$	$e^{\frac{2\pi i (n-1)}{n}}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
χ_{ϕ_k}	1	$e^{\frac{2\pi i k}{n}}$	$\dots$	$e^{\frac{2\pi i k u}{n}}$	$\dots$	$e^{\frac{2\pi i k (n-1)}{n}}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
$\chi_{\phi_{n-1}}$	1	$e^{\frac{2\pi i (n-1)}{n}}$	$\dots$	$e^{\frac{2\pi i (n-1) u}{n}}$	$\dots$	$e^{\frac{2\pi i (n-1)(n-1)}{n}}$

We will also compute the real representations of C_n .

Lemma 4.2.3 Let $A \in GL(d, \mathbb{C})$ be a matrix such that $A^n = I$ for some $n \in \mathbb{N}$. Then there is a matrix $Q \in GL(d, \mathbb{C})$ such that

$$Q^{-1}AQ = \begin{pmatrix} \theta_1 & & \\ & \ddots & \\ & & \theta_d \end{pmatrix}$$

where $\theta_1, \dots, \theta_d$ are n th roots of unity and the matrix is everywhere else 0.

Proof Let $f(X) = X^n - 1$. Then Clearly $f(A) = 0$. This means that the minimal polynomial $\mu_A(X)$ of A divides $f(X) = X^n - 1$. The roots of f are the n -roots of 1, namely $1, \zeta, \dots, \zeta^{n-1}$ where $\zeta = e^{2\pi i/n}$. Since $\mu_A(X)$ divides $f(X)$, the roots of μ_A are the n -roots of 1. Moreover, $f(X) = X^n - 1$ has distinct roots, and so μ_A has distinct roots. Hence we know that A is diagonalizable with entries the n -roots of unity. ■

Theorem 4.2.4 Denote $C_n = \langle x \rangle$ the cyclic group. Let $\rho : C_n \rightarrow GL(d, \mathbb{C})$ be a representation. Then there exists $\theta_1, \dots, \theta_d$ which are n th roots of unity such that the representation $\rho' : C_n \rightarrow GL(d, \mathbb{C})$ defined by

$$\rho'(x^k) = \begin{pmatrix} \theta_1^k & & \\ & \ddots & \\ & & \theta_d^k \end{pmatrix}$$

is equivalent to ρ .

Proof Let $A = \rho(x)$. Since $x^n = 1$ and ρ is a homomorphism we have $A^n = \rho(x^n) = \rho(1) = 1$. By the above lemma there exists $Q \in GL(d, \mathbb{C})$ and $\theta_1, \dots, \theta_d$ n th roots of unity such that $Q^{-1}\rho(x)Q = \text{diag}(\theta_1, \dots, \theta_d)$. Define $\rho' : C_n \rightarrow GL(d, \mathbb{C})$ by

$$\rho'(x^k) = Q^{-1}\rho(x)Q$$

This is a representation equivalent to ρ by lemma 1.1.2. Finally we have that

$$\begin{aligned} \rho'(x^k) &= Q^{-1}\rho(x^k)Q \\ &= Q^{-1}A^kQ \\ &= (Q^{-1}AQ)^k \\ &= \begin{pmatrix} \theta_1^k & & \\ & \ddots & \\ & & \theta_d^k \end{pmatrix} \end{aligned}$$

Thus we are done. ■

4.3 Representations of the Quaternion Group

Theorem 4.3.1 The one dimensional complex representations of

$$Q_8 = \langle a, b \mid a^4, a^2b^{-2}, abab^{-1} \rangle$$

are given as follows. Let $\psi_{(u,v)} : Q_8 \rightarrow \mathbb{C}^*$ be map sending a to u and b to v . Then the full list of one dimensional complex representations is given by

$$\{\psi_{(1,1)}, \psi_{(1,-1)}, \psi_{(-1,1)}, \psi_{(-1,-1)}\}$$

Proof By the universal property of the free group, if $\psi : Q_8 \rightarrow \mathbb{C}^*$ is a group homomorphism, then $\phi(a)^4 = 1$, $\phi(a)^2 = \phi(b)^2$ and $\psi(b)\psi(a)\psi(b)^{-1} = \psi(a)^{-1}$ which implies that $\psi(a) = \psi(a)^{-1}$ since $\mathbb{C}^*$ is abelian. It is then clear that $\psi(a)^4 = \psi(b)^4 = 1$ implies that $\psi(a), \psi(b) = 1, i, -1$ or $-i$.

If $\psi(a) = 1$, then $\psi(b) = 1$ gives the trivial representation. Using the given relations, we deduce that $\psi(b)^2 = 1$ hence specifying that $\psi(b) = -1$ gives another representation.

If $\psi(a) = -1$, then $\psi(b)^2 = 1$ hence $\psi(b) = \pm 1$. Note that the final relation is also satisfied. Hence this gives two possible representations.

If $\psi(a) = \pm i$, then $\psi(b) = \pm i$. But $i^{-1} = -i$ hence the last relation cannot be satisfied. Thus there are no representations coming from this case. Thus we conclude. ■

Proposition 4.3.2 Write the quaternion group with the following free group presentation:

$$Q_8 = \langle a, b \mid a^4, a^2b^{-2}, abab^{-1} \rangle$$

There is a two dimensional complex irreducible representation $\phi : Q_8 \rightarrow \text{GL}(2, \mathbb{C})$ given by

$$\phi(a) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad \text{and} \quad \phi(b) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Proof By the universal property of the free group with relations, we just have to show that $\phi(a)^4 = I$, $\phi(a)^2 = \phi(b)^2$ and $\phi(a)\phi(b)\phi(a)\phi(b)^{-1} = I$. We have that

$$\phi(a)^4 = \begin{pmatrix} i^4 & 0 \\ 0 & (-i)^4 \end{pmatrix} = I$$

Thus the first relation is satisfied. For the second relation, we have that

$$\phi(a)^2 = \begin{pmatrix} i^2 & 0 \\ 0 & (-i)^2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

and

$$\phi(b)^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Thus the second relation is satisfied. For the third relation, we have that

$$\begin{aligned}
 \phi(a)\phi(b)\phi(a)\phi(b)^{-1} &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1} \\
 &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \\
 &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \\
 &= I
 \end{aligned}$$

We conclude that ϕ is a group homomorphism. It is irreducible since $\phi(a)$ and $\phi(b)$ has no common eigenvector and hence no subspace is preserved by both $\phi(a)$ and $\phi(b)$. $\blacksquare$

Theorem 4.3.3 The complete list of complex irreducible representations of $Q_8 = \langle a, b \mid a^4, a^2b^{-2}, abab^{-1} \rangle$ is given by

$$\{\psi_{(1,1)}, \psi_{(1,-1)}, \psi_{(-1,1)}, \psi_{(-1,-1)}, \phi\}$$

Proof We have that

$$\dim(\psi_{(1,1)}) + \dim(\psi_{(1,-1)}) + \dim(\psi_{(-1,1)}) + \dim(\psi_{(-1,-1)}) + \dim(\phi) = 4 + 4 = 8 = |Q_8|$$

By 2.4.3, we conclude. $\blacksquare$

Proposition 4.3.4 There are 5 conjugacy classes of Q_8 and is given as follows:

$$\{1\}, \{a, a^3\}, \{ab, a^3b\}, \{b, a^2b\}, \{a^2\}$$

Theorem 4.3.5 The character table of Q_8 is given as follows.

G	1	$\{a, a^3\}$	$\{ab, a^3b\}$	$\{b, a^2b\}$	$\{a^2\}$
$\chi_{\psi_{(1,1)}}$	1	1	1	1	$\dots$
$\chi_{\psi_{(1,-1)}}$	1	1	-1	-1	1
$\chi_{\psi_{(-1,1)}}$	1	-1	-1	1	1
$\chi_{\psi_{(-1,-1)}}$	1	-1	1	-1	1
χ_ϕ	2	0	0	0	-2

4.4 Representations of the Dihedral Group

Let us first compute the one dimensional complex representations of D_{2n} .

Theorem 4.4.1 Let $n \in \mathbb{N}$. The one dimensional complex representations of

$$D_{2n} = \langle r, s \mid r^n, s^2, rs = sr^{-1} \rangle$$

are given as follows. Let $\phi_{(u,v)} : D_{2n} \rightarrow \mathbb{C}^*$ be the map sending r to u and s to v .

- If n is odd, then the full list of representations is given by

$$\{\phi_{(1,1)}, \phi_{(1,-1)}\}$$

- If n is even, then the full list of representations is given by

$$\{\phi_{(1,1)}, \phi_{(1,-1)}, \phi_{(-1,1)}, \phi_{(-1,-1)}\}$$

Proof By the universal property of a free group with relations, any group homomorphism $\phi : D_{2n} \rightarrow \mathbb{C}^*$ is determined by the elements r and s such that $\phi(r)^n = 1$ and $\phi(s)^2 = 1$ and $\phi(r)\phi(s) = \phi(s)\phi(r)^{-1}$. Since $\mathbb{C}^*$ is abelian, the last condition becomes $\phi(r) = \phi(r)^{-1}$ which means that $\phi(r) = \pm 1$. Similarly, we have $\phi(s) = \pm 1$. When n is even, we precisely have four non-equivalent representations. When n is odd, then $(-1)^n \neq 1$ so $\phi(r)^n = 1$ is no longer satisfied if we choose $\phi(r) = -1$. This leaves only two choices. Thus we conclude. ■

Proposition 4.4.2 Let $n \in \mathbb{N}$. Write $D_{2n} = \langle r, s \mid r^n, s^2, rs = sr^{-1} \rangle$ the dihedral group. For $h \in \mathbb{Z}$, define $\rho^h : D_{2n} \rightarrow \text{GL}(2, \mathbb{C})$ by

$$\rho^h(r^k) = \begin{pmatrix} \zeta^{hk} & 0 \\ 0 & \zeta^{-hk} \end{pmatrix} \quad \text{and} \quad \rho^h(sr^k) = \begin{pmatrix} 0 & \zeta^{hk} \\ \zeta^{-hk} & 0 \end{pmatrix}$$

where $\zeta = e^{2\pi i/n}$. Then the following are true.

- ρ^h is a representation of D_{2n}
- ρ^h is equivalent to $\text{Ind}_{C_n}^{D_{2n}} \rho_{\phi_h}$ where ϕ_h is as in theorem 4.2.1.
- $\rho^h = \rho^{h+n}$
- ρ^h is equivalent to ρ^{n-h}

Proof It is clear that ρ^h is a representation since the following are true.

- $(\rho^h(r))^n = \begin{pmatrix} \zeta^h & 0 \\ 0 & \zeta^{-h} \end{pmatrix}^n = \begin{pmatrix} \zeta^{hn} & 0 \\ 0 & \zeta^{-hn} \end{pmatrix} = I$
- $(\rho^h(s))^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = I$
- $\rho^h(s)\rho^h(r)\rho^h(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \zeta^h & 0 \\ 0 & \zeta^{-h} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \zeta^{-h} & 0 \\ 0 & \zeta^h \end{pmatrix} = (\rho^h(r))^{-1}$

We have by definition of the induced representation that

$$\text{Ind}_{C_n}^{D_{2n}} \phi_k(r^t) = \begin{pmatrix} \phi_k(r^t) & \phi_k(r^t s) \\ \phi_k(sr^t) & \phi_k(sr^t s) \end{pmatrix} = \begin{pmatrix} \phi_k(r^t) & \phi_k(r^t s) \\ \phi_k(sr^t) & \phi_k(r^{-t}) \end{pmatrix} = \begin{pmatrix} \zeta^{kt} & 0 \\ 0 & \zeta^{-kt} \end{pmatrix}$$

and we have that

$$\text{Ind}_{C_n}^{D_{2n}} \phi_k(sr^t) = \begin{pmatrix} \phi_k(sr^t) & \phi_k(sr^t s) \\ \phi_k(r^t) & \phi_k(r^t s) \end{pmatrix} = \begin{pmatrix} \phi_k(sr^t) & \phi_k(r^{-t}) \\ \phi_k(r^t) & \phi_k(r^t s) \end{pmatrix} = \begin{pmatrix} 0 & \zeta^{-kt} \\ \zeta^{kt} & 0 \end{pmatrix}$$

But $\begin{pmatrix} 0 & \zeta^{-kt} \\ \zeta^{kt} & 0 \end{pmatrix}$ is similar to $\begin{pmatrix} 0 & \zeta^{kt} \\ \zeta^{-kt} & 0 \end{pmatrix}$ via the invertible matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hence $\text{Ind}_{C_n}^{D_{2n}} \phi_k$ is equivalent to ρ^k .

For the third item, notice that ζ is the n th root of unity so $\zeta^{hk} = \zeta^{(h+n)k}$ and hence $\rho^h = \rho^{h+n}$.

Finally, notice that $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is such that $\rho^h(x)$ and $\rho^{n-h}(x)$ to be similar matrices for any $x \in D_{2n}$.

Indeed we have that

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \rho^{n-h}(r^k) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \zeta^{-(n-h)k} & 0 \\ 0 & \zeta^{(n-h)k} \end{pmatrix} \sim \begin{pmatrix} \zeta^{hk} & 0 \\ 0 & \zeta^{-hk} \end{pmatrix} = \rho^h(r^k)$$

and

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \rho^{n-h}(sr^k) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \zeta^{-(n-h)k} \\ \zeta^{(n-h)k} & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & \zeta^{hk} \\ \zeta^{-hk} & 0 \end{pmatrix} = \rho^h(sr^k)$$

■

The above proposition shows that the only meaningful 2-dimensional representations of D_{2n} of the above form is ρ^h for $0 \leq h \leq n/2$.

Proposition 4.4.3 Let $n \in \mathbb{N}$. Write $D_{2n} = \langle r, s \mid r^n, s^2, rs = sr^{-1} \rangle$ the dihedral group. For $0 < h < n/2$, define $\rho^h : D_{2n} \rightarrow \text{GL}(2, \mathbb{C})$ by

$$\rho^h(r^k) = \begin{pmatrix} \zeta^{hk} & 0 \\ 0 & \zeta^{-hk} \end{pmatrix} \quad \text{and} \quad \rho^h(sr^k) = \begin{pmatrix} 0 & \zeta^{hk} \\ \zeta^{-hk} & 0 \end{pmatrix}$$

where $\zeta = e^{2\pi i/n}$. Then all the ρ^h for $0 < h < n/2$ are pairwise non-equivalent and irreducible.

Proof Notice that $\rho^h(r)$ stretches the axis thus the eigenspaces of $\rho^h(r)$ are the coordinate axes. But these are not stable under the complex reflection $\rho^h(s)$. Thus there are no subspaces of $\mathbb{C}^2$ that is fixed by ρ^h . Thus ρ^h is irreducible. Now the characters for χ_{ρ^h} is given by

$$\chi_{\rho^h}(r^k) = \zeta^{hk} + \zeta^{-hk} = 2 \cos\left(\frac{2\pi hk}{n}\right)$$

and $\chi_{\rho^h}(sr^k) = 0$. But χ is a complete invariant of representations hence for $0 < h \neq t < n/2$, $\chi_{\rho^h}(r^k) \neq \chi_{\rho^t}(r^k)$ implies $\chi_{\rho^h} \neq \chi_{\rho^t}$ implies ρ^h and ρ^t are not equivalent. ■

Theorem 4.4.4 Let $n \in \mathbb{N}$. Then the group $D_{2n} = \langle r, s \mid r^n, s^2, rs = sr^{-1} \rangle$ has the following complete list of irreducible complex representations.

- If n is odd, then the full list is given by

$$\{\phi_{(1,1)}, \phi_{(1,-1)}\} \cup \{\rho^h \mid 0 < h < \frac{n}{2}\}$$

- If n is even, then the full list is given by

$$\{\phi_{(1,1)}, \phi_{(1,-1)}, \phi_{(-1,1)}, \phi_{(-1,-1)}\} \cup \{\rho^h \mid 0 < h < \frac{n}{2}\}$$

Proof We have seen that the above lists consists only of irreducible representations. When n is odd, we have that

$$\dim(\phi_{(1,1)}) + \dim(\phi_{(1,-1)}) + \sum_{h=1}^{(n-1)/2} \dim(\rho^h) = 1 + 1 + \left(\frac{n-1}{2}\right) \cdot 4 = 2n$$

By lemma 2.4.3, we conclude that the given list is the complete list of irreducible complex representations for D_{2n} when n is odd.

When n is even, we have that

$$\dim(\phi_{(1,1)}) + \dim(\phi_{(1,-1)}) + \dim(\phi_{(-1,1)}) + \dim(\phi_{(-1,-1)}) + \sum_{h=1}^{n/2-1} \dim(\rho^h) = 1 + 1 + 1 + 1 + \left(\frac{n}{2} - 1\right) \cdot 4$$

which is equal to $2n$. By lemma 2.4.3, we conclude that the given list is the complete list of irreducible complex representations for D_{2n} when n is even. ■

5 Some Combinatorics

5.1 Partitions

(To be separated to notes on combinatorics:)

Definition 5.1.1 (Partitions of an Integer) Let $n \in \mathbb{N}$. A partition λ of n is a list $(\lambda_1, \lambda_2, \dots, \lambda_l)$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_l > 0$ and $\sum_{i=1}^l \lambda_i = n$.

Definition 5.1.2 (Dominance of Partitions) Let λ and μ be two partitions of n . We say that λ dominates μ if for all $k \in \mathbb{N}$,

$$\sum_{i=1}^k \lambda_i \geq \sum_{i=1}^k \mu_i$$

In this case we denote it as $\lambda \supseteq \mu$.

Lemma 5.1.3 Let $n \in \mathbb{N}$. Then the set of partitions of n together with dominance forms a poset.

Definition 5.1.4 (Lexicographical Ordering) Let λ and μ be partitions of n . We say that $\lambda < \mu$ if for some $i \in \mathbb{N}$, the following are true.

- For all $j < i$, $\lambda_j = \mu_j$
- $\lambda_i < \mu_i$

Lemma 5.1.5 Let $n \in \mathbb{N}$. Then the lexicographical order on the set of all partitions of n form a total ordering.

Proposition 5.1.6 Let λ and μ be partitions of n . If $\lambda \supseteq \mu$, then $\lambda \geq \mu$.

5.2 Young Diagrams and Young Tabloids

Definition 5.2.1 (Young Diagram) Let λ be a partition of n . The young diagram of λ is a collection of boxes arranged in left-justified rows, such that row i has λ_i many boxes.

Definition 5.2.2 (Young Tableau) Let λ be a partition of n . A Young tableau is obtained by filling in the boxes of the Young diagram of λ with the numbers $1, \dots, n$. It is also called a λ -tableau.

Definition 5.2.3 (Tabloids) Let λ be a partition of n . Let t_1 and t_2 be two Young tableaux. We say that t_1 and t_2 are row equivalent if for all i , the i th row of t_1 and t_2 contain the same element. A λ -tabloid is an equivalence class of λ -tableau.

Proposition 5.2.4 Let λ be a partition of n . Let T be the set of all λ -tabloids. Then S_n acts on T by

$$\tau \cdot [t_j] = [\tau(t_j)]$$

Definition 5.2.5 (Young Subgroup) Let λ be a partition of n . Define the Young subgroup of $S_n \leq S_n$ to λ is defined as

$$S_\lambda = S_{1, \dots, \lambda_1} \times S_{\lambda_1+1, \dots, \lambda_1+\lambda_2} \times \dots \times S_{n-\lambda_l+1, \dots, n}$$

Definition 5.2.6 (Row and Column Stabilizers) Let t be a tableau with rows $R_1, \dots, R_l$ and columns $C_1, \dots, C_k$. Define the row stabilizer of t to be the group

$$R_t = S_{R_1} \times \dots \times S_{R_l}$$

Define the column stabilizer of t to be the group

$$C_t = S_{C_1} \times \cdots \times S_{C_k}$$

For $\tau \in R_t$, we have that $\tau([t]) = [t]$. In fact every element of $[t]$ is given in this way. Thus by writing

$$R_t t = \{\tau(t) \mid \tau \in R_t\}$$

we can identify the tabloid as

$$[t] = R_t t$$

6 Representations of the Symmetric Group

6.1 Representations Associated to a Partition

Recall that a partition $\lambda = (\lambda_1, \dots, \lambda_k)$ of an integer n gives a Young diagram, which n left-aligned boxes with λ_i boxes on row i . By writing $1, \dots, n$ in the boxes, we obtain a Young tableau. We then consider its equivalence class of Young tableau, in which we say that two Young tableau are equivalent if row i of the two tableau contains the same element for all i . Such an equivalence class is called a Young tabloid of shape λ .

We have also seen that S_n acts on the set of all tabloids by permuting the numbers in the boxes. The stabilizer of a tabloid t is precisely the row stabilizers $R_t = S_{R_1} \times \dots \times S_{R_l}$, for $R_1, \dots, R_l$ the rows of the tabloid.

Definition 6.1.1 (Permutation Module Associated to a Partition) Let λ be a partition of n . Let $[t_1], \dots, [t_k]$ be a complete list of λ -tabloids. Define the permutation module associated to λ to be

$$M^\lambda = \mathbb{C}\langle [t_1], \dots, [t_k] \rangle$$

This means that $\rho : S_n \rightarrow \text{GL}(M^\lambda)$ where

$$\rho(\tau) : M^\lambda \rightarrow M^\lambda$$

for $\tau \in S_n$ is defined on the basis by sending $[t_j]$ to $[\tau(t_j)]$.

Proposition 6.1.2 Let λ be a partition of n . Let $T = S_n/S_\lambda$ be the set of cosets of S_λ . Then the following representations are equivalent.

- The induced representation $\text{Ind}_{S_\lambda}^{S_n} 1$
- The permutation representation $\rho : S_n \rightarrow \text{GL}(CT)$ where $\rho(\tau) : CT \rightarrow CT$ for $\tau \in S_n$ is defined on the basis by sending $\pi_k S_\lambda$ to $\tau \pi_k S_\lambda$
- The permutation module $\psi : S_n \rightarrow \text{GL}(M^\lambda)$ corresponding to λ

Recall that for a tableau t , the columns stabilizer of t is given by

$$C_t = S_{C_1} \times \dots \times S_{C_k}$$

if $C_1, \dots, C_k$ are the columns of the tableau.

Definition 6.1.3 (Associated Polytabloid) Let λ be a partition of n . Let t be a λ -tableau. Define the associated polytabloid $e_t \in M^\lambda$ by

$$e_t = \sum_{\pi \in C_t} \text{sign}(\pi)(\pi \cdot [t])$$

Lemma 6.1.4 Let t be a tableau of n boxes and let $\pi \in S_n$ be a permutation. Then the following are true.

- $R_{\pi t} = \pi R_t \pi^{-1}$
- $C_{\pi t} = \pi C_t \pi^{-1}$
- $\kappa_{\pi t} = \pi \kappa_t \pi^{-1}$
- $e_{\pi t} = \pi e_t$.

6.2 Representations of the Symmetric Group

Definition 6.2.1 (Specht Module) Let λ be a partition of n . Define the Specht module S^λ to be the submodule of M^λ where

$$S^\lambda = \mathbb{C}\langle e_t \mid t \text{ has shape } \lambda \rangle$$

Lemma 6.2.2 Let $n \in \mathbb{N}$. Then the following are true

- Then the representation $\rho : S_n \rightarrow \text{GL}(S^{(n)})$ is the trivial representation.
- The representation $\rho : S_n \rightarrow \text{GL}(S^{(1^n)})$ is equivalent to the sign representation.

Lemma 6.2.3 Let λ and μ be partitions of n . Let t be a λ -tableau and let s be a μ -tableau. Then the following are true.

- If $\sum_{\pi \in C_t} \text{sign}(\pi)(\pi \cdot [s]) \neq 0$, then $\lambda \supseteq \mu$.
- If $\lambda = \mu$, then $\sum_{\pi \in C_t} \text{sign}(\pi)(\pi \cdot [s]) \in \{-e_t, e_t\}$.

Theorem 6.2.4 (The Submodule Theorem) Let λ be a partition of n . Let U be a submodule of M^λ . Then either $S^\lambda \subseteq U$ or $U \subseteq (S^\lambda)^\perp$.

Lemma 6.2.5 Let λ and μ be partitions of n . Let $f \in \text{Hom}_{S_n}(S^\lambda, M^\mu)$ be non-zero. Then $\lambda \supseteq \mu$. If $\lambda = \mu$ then f is multiplication by a scalar.

Proof S^λ is generated by e_t by definition. Since $f \neq 0$, there exists a tableau t such that $f(e_t) \neq 0$. Since S^λ is a submodule of M^λ , we have that $M^\lambda = S^\lambda \oplus (S^\lambda)^\perp$. Define an extension $\tilde{f} : M^\lambda \rightarrow M^\mu$ of f as follows. For $v \in (S^\lambda)^\perp$, set $\tilde{f}(v) = 0$. Now we have that ??? (5.4.2) ■

Theorem 6.2.6 Let $n \in \mathbb{N}$. Then the set of all S^λ for λ a partition of n forms a complete list of irreducible S_n -representations.

Proof We first prove that S^λ is irreducible. Let $U \subseteq S^\lambda$ be a subrepresentation. By the submodule theorem, either $S^\lambda \subseteq U$ or $U \subseteq (S^\lambda)^\perp$. Thus $S^\lambda = U$ or $U \subseteq S^\lambda \cap (S^\lambda)^\perp = 0$.

It is clear that the number of conjugacy classes of S_n is equal to the number of partitions of n . Thus the set of all S^λ for λ a partition of n has cardinality equal to the the number of conjugacy classes of S_n . By theorem 2.4.6, the number of pairwise non-isomorphic irreducible representations of S_n is equal to the number of conjugacy classes of S_n . Thus it remains to show that they are pairwise non-isomorphic.

Suppose that S^λ is equivalent to S^μ for some partitions λ and μ of n . Then there exists a non-zero $f \in \text{Hom}_{S_n}(S^\lambda, S^\mu)$ which can be interpreted as an element of $\text{Hom}_{S_n}(S^\lambda, M^\mu)$. By the above lemma, we conclude that $\lambda \supseteq \mu$. Similarly, we conclude that $\mu \supseteq \lambda$. Thus $\lambda = \mu$. ■

6.3 Character Tables for S_3

Proposition 6.3.1 (The Sign Representation) Let $n \in \mathbb{N}$. Then the map $\rho : S_n \rightarrow \mathbb{C}^*$ defined by

$$\rho(\tau) = \text{sign}(\tau)$$

for $\tau \in S_n$ is a one dimensional complex representation of S_n .

Recall that any two permutations in S_n of the same cycle type are conjugate. This completely determines the conjugacy classes of S_n . In particular, this means that the number of conjugacy classes of S_n is given by the number of integer partitions of n .

Proposition 6.3.2 Let $X = \{a, b, c\}$ such that S_3 acts on X by permutation. Consider the permutation representation $\rho : S_3 \rightarrow \text{GL}(\mathbb{C}X)$. Then the subspace

$$U = \mathbb{C}\langle a + b + c \rangle$$

of $\mathbb{C}X$ is a subrepresentation of G .

Proof We want to show that $\tau u \cdot u \in U$ for all $u \in U$. We just have to check this for the basis. We have that

$$\tau u \cdot (a + b + c) = \tau(a) + \tau(b) + \tau(c)$$

Since τ is a permutation, $\{\tau(a), \tau(b), \tau(c)\} = \{a, b, c\}$ hence we are done. ■

Proposition 6.3.3 Let $X = \{a, b, c\}$ such that S_3 acts on X by permutation. Consider the permutation representation $\rho : S_3 \rightarrow \text{GL}(\mathbb{C}X)$ and the subrepresentation $U = \mathbb{C}\langle a + b + c \rangle$ of $\mathbb{C}X$. Then V/U is an irreducible representation of G .

Proof We already know that V/U is a representation, it remains to show that it is irreducible. We wish to invoke 2.3.5. The representation $\bar{\rho} : S_3 \rightarrow \text{GL}(V/U)$ is given by $\rho(\tau)(v + U) = \tau(v) + U$. We give a non-canonical basis of $\{a + V/U, b + V/U\}$ to V/U . The character of V/U is given as follows.

There are three conjugacy classes of S_3 given by $\{1\}$, $\{(a, b), (b, c), (a, c)\}$ and $\{(a, b, c), (a, c, b)\}$. We have that

- $\chi_{\bar{\rho}}(1) = \text{tr}(I) = 2$
- Under the given basis, the matrix of $\bar{\rho}(a, b)$ is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Hence $\chi_{\bar{\rho}}((a, b)) = 0$
- Under the given basis, the matrix of $\bar{\rho}(a, b, c)$ is $\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. Hence $\chi_{\bar{\rho}}((a, b, c)) = -1$

Now we have that

$$\begin{aligned} \langle \chi_{\bar{\rho}}, \chi_{\bar{\rho}} \rangle &= \frac{1}{|S_3|} \sum_{\tau \in S_3} \chi_{\bar{\rho}}(\tau)^2 \\ &= \frac{1}{6} (1 \cdot 2^2 + 3 \cdot 0 + 2 \cdot (-1)^2) \\ &= 1 \end{aligned}$$

By lemma 2.3.5, we conclude that V/U is irreducible. ■

Theorem 6.3.4 There are a total of three irreducible complex representations of S_3 given by

- The trivial representation $\rho_1(\tau) = 1$
- The sign representation $\rho_2(\tau) = \text{sign}(\tau)$
- The representation $\bar{\rho}$ given by 6.3.3.

Proof We have seen that each of these are irreducible. There are three conjugacy classes of S_3 . By 2.4.6 we conclude that we have exhausted all the irreducible representations. ■

Theorem 6.3.5 The character table of S_3 is given as follows.

G	1	$\{(1, 2), (1, 3), (2, 3)\}$	$\{(1, 2, 3), (1, 3, 2)\}$
χ_{trivial}	1	1	1
χ_{sign}	1	-1	1
$\chi_{\bar{\rho}}$	2	0	-1

6.4 Representations of the Alternating Group